CGS^{3}GCN: Center-Pixel Gated Spatial–Spectral Spiking Graph Convolution Network for Hyperspectral Image Classification

Shufang Xu, *Member, IEEE*, Wenxuan Fei, Shuyu Fei, and Hongmin Gao, *Senior Member, IEEE*

***Abstract*—Hyperspectral image classification (HSIC) remains challenging due to complex spatial–spectral dependencies, severe spectral redundancy, and the difficulty of learning discriminative representations from high-dimensional hyperspectral data. Graph convolutional networks (GCNs) provide an effective framework for modeling non-Euclidean spatial–spectral relationships, while spiking neural networks (SNNs) enable robust spike-based representation learning. However, existing graph-based HSIC methods mainly emphasize spatial graph modeling, whereas effective spectral graph learning remains underexplored. Moreover, the temporal dynamics encoded in spike sequences are not fully utilized for discriminative representation learning. To address these issues, we propose a center-pixel gated spatial–spectral spiking graph convolution network (CGS^{3}GCN). Specifically, a prototype-conditioned spectral embedding graph (PCSEG) is designed, where spectral embedding channels are treated as graph nodes and prototype-conditioned descriptors are used to construct node representations, enabling compact and discriminative spectral dependency modeling. A spike temporal–channel attention (STCA) module is developed to exploit spike-based global statistics (SGS) and cross-timestep channel interactions for adaptive feature recalibration. Furthermore, a center-pixel gate (CPG) is introduced to refine graph propagation according to center-pixel semantics. Finally, spatial and spectral graph representations are adaptively fused for classification. Experimental results on four benchmark datasets demonstrate that CGS^{3}GCN achieves state-of-the-art (SOTA) classification performance. In addition, it exhibits strong robustness under limited training samples and noisy conditions. The source code will be released at: https://github.com/fwx0314/CGS3GCN**

***Index Terms*—Center-pixel gating, graph convolutional networks (GCNs), hyperspectral image classification (HSIC), spectral graph learning, spiking neural networks (SNNs).**

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Shufang Xu, Wenxuan Fei, and Shuyu Fei are with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China (e-mail: xushufang@hhu.edu.cn; 241607010003@hhu.edu.cn; feishuyu@hhu.edu.cn).

Hongmin Gao is with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China, also with the School of Information Engineering, Xinjiang Institute of Technology, Aksu 843000, China, and also with the School of Information and Design, Zhejiang Industry Polytechnic College, Shaoxing 312000, China (e-mail: gaohongmin@hhu.edu.cn).

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I. Introduction

HYPERSPECTRAL images (HSIs) contain hundreds of contiguous spectral bands, providing rich spectral–spatial information for accurate material discrimination [1]. Benefiting from their high spectral resolution, HSIs have been widely applied in remote sensing [2], agriculture [3], environmental monitoring [4], and medical diagnosis [5]. However, the high dimensionality, spectral redundancy, and noise sensitivity of hyperspectral data pose significant challenges for effective representation and classification [6], [7].

As a fundamental task in hyperspectral analysis, HSI classification (HSIC) aims to assign semantic labels to pixels according to their spectral–spatial characteristics. Early HSIC methods mainly relied on handcrafted features and conventional machine learning techniques, such as support vector machines (SVMs) [8] and random forests (RFs) [9]. Although these methods achieved promising performance under limited training samples, their capability to model complex nonlinear spectral–spatial relationships remains limited [10].

With the development of deep learning, convolutional neural networks (CNNs) have been widely introduced into HSIC to jointly model spectral–spatial features [11], [12]. However, their limited receptive fields restrict the capture of long-range contextual dependencies [13]. To address this, Transformer-based methods have demonstrated strong global modeling capability through the self-attention mechanism [14], [15]. Despite their effectiveness, reshaping hyperspectral features into token sequences may weaken intrinsic spatial structures [16], and the quadratic complexity of self-attention introduces considerable computational and memory overhead [17].

Graph convolutional networks (GCNs) have emerged as an effective solution for HSIC due to their strong non-Euclidean representation capability [18], [19]. By modeling graph-structured relationships, GCN-based methods can effectively capture irregular spatial structures and long-range contextual dependencies that are difficult to characterize using conventional convolution operations [20], [21]. Moreover, graph representations provide a flexible framework for jointly modeling spatial and spectral correlations in hyperspectral data, making GCNs particularly suitable for hyperspectral representation learning.

Recently, spiking neural networks (SNNs) have attracted increasing attention in HSIC due to their event-driven information processing mechanism and inherent robustness to noise perturbations [22], [23]. By representing information as sparse

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spike sequences, SNNs naturally suppress continuous noise interference through discrete firing activities, making them particularly suitable for hyperspectral data characterized by spectral variability and noise contamination [24]. Moreover, the spike generation process produces rich temporal firing patterns, which provide additional representation capacity beyond conventional static neural networks. These properties make SNNs a promising framework for HSIC.

Consequently, integrating the non-Euclidean representation capability of GCNs with the temporal spike dynamics of SNNs has emerged as a promising paradigm for discriminative hyperspectral representation learning. However, developing an effective spatial–spectral spiking graph framework for HSIC still faces three major challenges.

1) Existing graph-based HSIC methods mainly focus on spatial graph modeling, while spectral graph learning remains relatively underexplored. Existing spectral graphs typically construct node representations directly from channelwise spatial responses, which contain redundant information and lack discriminative semantic cues.
2) Although SNNs provide rich temporal firing patterns through spike generation, most existing SNN-based HSIC methods primarily use spike responses for feature propagation, while the temporal statistics and cross-timestep interactions embedded in spike sequences remain insufficiently explored.
3) In patch-based HSIC, the classification label is determined by the center pixel, whose spatial–spectral signature embodies the class-defining semantic information. However, existing graph propagation indiscriminately aggregates neighboring features—at category boundaries, heterogeneous neighbors contaminate the target pixel's class-specific representation. A parallel issue arises in spectral propagation, where interchannel mixing dilutes target-related spectral semantics.

To address these issues, we propose a center-pixel gated spatial–spectral spiking graph convolution network (CGS^{3}GCN) for HSIC. Specifically, hyperspectral features are first projected into a compact latent space and encoded into spike sequences. A spike temporal–channel attention (STCA) module is then employed to exploit spike-based global statistics (SGS) and cross-timestep channel modeling (CT) for adaptive temporal–channel feature recalibration. Based on the recalibrated spike features, two complementary graph branches are constructed. For spectral modeling, a prototype-conditioned spectral embedding graph (PCSEG) is proposed. Unlike conventional spectral graphs that construct node representations directly from raw channelwise spatial responses—which are often redundant and semantically uninformative—PCSEG conditions each spectral embedding node on complementary prototypes extracted from the same hyperspectral patch. Specifically, a center prototype preserves target-aware semantics directly associated with the classification label, a mean prototype captures the overall contextual spectral distribution, and a max prototype emphasizes dominant salient spectral responses. These three prototypes collectively provide target-aware, contextual, and salient perspectives of the spectral distribution. By conditioning node representations on these semantically complementary prototypes rather than on raw feature responses, PCSEG produces compact, semantically enriched node descriptors. Spectral embedding channels are then treated as graph nodes, and spectral dependencies are modeled in this prototype-conditioned latent embedding space, enabling more discriminative and compact spectral graph learning. For spatial modeling, an adaptive spatial graph is built to characterize non-Euclidean contextual relationships among neighboring pixels. To preserve discriminative information during graph propagation, a center-pixel gate (CPG) is introduced in both graph branches to adaptively refine graph representations according to the semantic relevance of the center pixel. Finally, the spatial and spectral graph representations are adaptively fused for classification.

The main contributions are summarized as follows.

1) We propose a PCSEG, which constructs prototype-conditioned node representations for spectral embedding channels to enable compact and discriminative spectral dependency modeling.
2) We develop an STCA mechanism that jointly exploits SGS and cross-timestep channel dependencies to enhance temporal–channel feature discrimination in SNN-based HSIC.
3) We introduce a CPG that preserves target-oriented semantics during graph propagation by using the center-pixel representation as adaptive semantic guidance, thereby improving both spatial and spectral graph reasoning.
4) By integrating the above components within a unified spatial–spectral spiking graph learning framework, the proposed CGS^{3}GCN achieves state-of-the-art (SOTA) performance on multiple benchmark hyperspectral datasets and exhibits strong robustness under limited training samples and noisy conditions.

The remainder of this article is organized as follows. Section II reviews related studies on GCNs and SNNs for HSIC. Section III details the proposed CGS^{3}GCN framework. Section IV presents the experimental results and analysis. Finally, Section V concludes this article.

II. Related Works

This section briefly reviews recent studies on GCNs and SNNs for HSIC.

A. GCNs for HSIC

Graph-based representation learning has been widely adopted in HSIC to capture non-Euclidean spatial–spectral dependencies. Early spatial graph methods [25] relied on fixed global graphs, which were later improved by scalable minibatch training [26] and dynamic multiscale updating [27]. To further exploit complementary features, hybrid CNN-GCN architectures [18], [21], [28], [29] and attention-enhanced frameworks [20], [30] were proposed to integrate local contextual patterns with long-range graph reasoning. Parallel to spatial modeling, explicit spectral graphs [31], [32], [33] have emerged to capture interband relationships by treating channels or feature maps as graph nodes.

Despite these advances, most methods rely on superpixel-based aggregation or treat neighboring nodes equally during

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message passing, diluting target-pixel semantics in heterogeneous regions. Moreover, existing spectral graphs construct node representations directly from channelwise spatial responses, yielding redundant information and under-exploited spectral semantics.

B. SNNs for HSIC

Benefiting from their unique event-driven information processing and rich temporal dynamics, SNNs have been increasingly introduced into HSIC. Early works enabled gradient-based training via leaky integrate-and-fire (LIF) neurons and surrogate gradients [22], [34]. To enhance feature extraction, subsequent studies incorporated structural improvements such as residual connections [23], [35] and multibranch reparameterization [36]. In addition, various attention mechanisms—including temporal-channel [23], self-attention [37], and spatial–spectral attention [34]—have been integrated to capture critical spike dynamics. Recently, SNNs have also been hybridized with GANs [36], state-space models [38], and GCNs [39] to further boost long-range dependency modeling and representation capacity.

However, most SNN-based methods rely on direct spike responses for feature propagation, leaving temporal statistics and cross-timestep interactions in spike sequences underexplored. Furthermore, existing spiking graph frameworks rarely investigate explicit spectral graph learning and center-pixel semantic preservation, limiting their discriminability in complex scenes.

Building on our previous MS^{2}GCAN [39], the proposed CGS^{3}GCN further advances spiking graph learning by jointly integrating temporal–channel recalibration, prototype-conditioned spectral graph learning, and center-pixel semantic preservation, thereby addressing the limitations identified above.

III. Proposed Method

A. Preliminaries

In this work, we employ the discrete-time LIF neuron implemented via SpikingJelly [40]. The neuronal dynamics are formulated as

$$u[t] = v[t-1] - \beta\left(v[t-1] - V_{\text{reset}}\right) + X[t] \tag{1}$$

$$s[t] = \Theta\left(u[t] - \theta\right) \tag{2}$$

where $u[t]$ denotes the membrane potential before spike firing at timestep t, $v[t-1]$ denotes the membrane potential after reset at the previous timestep, β is the leakage factor, V_{reset} denotes the reset potential, $X[t]$ represents the synaptic input current, θ is the firing threshold, $\Theta(\cdot)$ denotes the Heaviside step function, and $s[t]$ represents the generated output spike. Specifically, the neuronal update proceeds in three stages. First, the membrane potential from the previous timestep $v[t-1]$ decays toward the reset potential V_{reset} with a rate controlled by the leakage factor β, while the incoming synaptic input $X[t]$ is accumulated, yielding the prespike membrane potential $u[t]$ in (1). Second, an output spike $s[t]$ is generated when $u[t]$ reaches or exceeds the firing threshold θ, as defined in (2). Finally, following spike generation, the membrane potential is hard-reset to V_{reset} for the next timestep.

To enable gradient-based optimization, the nondifferentiable spike activation function is approximated using the arctangent surrogate gradient

$$\frac{\partial s[t]}{\partial u[t]} \approx \frac{\alpha}{2\left(1 + \left(\frac{\pi}{2}\alpha\left(u[t] - \theta\right)\right)^2\right)} \tag{3}$$

where α controls the sharpness of the surrogate gradient approximation, and a larger α yields a closer approximation to the ideal spike derivative. For clarity, scalar formulations are presented here, while vectorized implementations are adopted throughout the framework.

For simplicity and structural clarity, the entire temporal dynamics and spike generation process of the LIF neuron across all timesteps are abstractly denoted as an operator

$$S = \text{LIF}(X) \tag{4}$$

where X and S represent the aggregated input current and the corresponding output spike sequence, respectively.

B. Overall Framework

Fig. 1 illustrates the overall architecture of the proposed CGS^{3}GCN, which follows a preprocessing–recalibration–dual-graph reasoning–adaptive fusion pipeline. Given an input HSI patch $\mathbf{X} \in \mathbb{R}^{L^2 \times S}$, it is first expanded across T timesteps and projected into a C-dimensional compact latent space to mitigate spectral redundancy. Since the HSI input is static, this expansion replicates the identical patch across all T timesteps; consequently, the temporal dynamics in the network arise entirely from the membrane integration and spike generation process of the LIF neurons, rather than from any temporal variation in the input representation. The STCA module then exploits SGS and cross-timestep channel dependencies to recalibrate temporal–channel representations, addressing the insufficient utilization of temporal statistics and intertimestep interactions in spike sequences. Based on the recalibrated features, two complementary graph branches are constructed. For spectral modeling, the PCSEG addresses a fundamental limitation of conventional spectral graphs: rather than constructing node representations directly from raw channelwise spatial responses that are redundant and semantically uninformative, PCSEG conditions each spectral embedding node on complementary center, mean, and max prototypes. These prototypes, respectively, distill the target-aware semantics, contextual statistics, and salient spectral responses from the same hyperspectral patch, producing semantically enriched and compact node descriptors. By modeling spectral dependencies among these prototype-conditioned embeddings, PCSEG facilitates more discriminative spectral graph learning while alleviating spectral redundancy. For spatial modeling, an adaptive spatial graph is built to characterize non-Euclidean contextual relationships among neighboring pixels. To prevent the dilution of center-pixel semantic information during graph propagation—where indiscriminate aggregation of heterogeneous neighboring features contaminates the class-defining spectral signature of the target pixel—a CPG is introduced in both branches. The CPG employs the center-pixel representation as adaptive semantic guidance to refine message aggregation, explicitly preserving target-oriented discriminative information throughout the propagation process.

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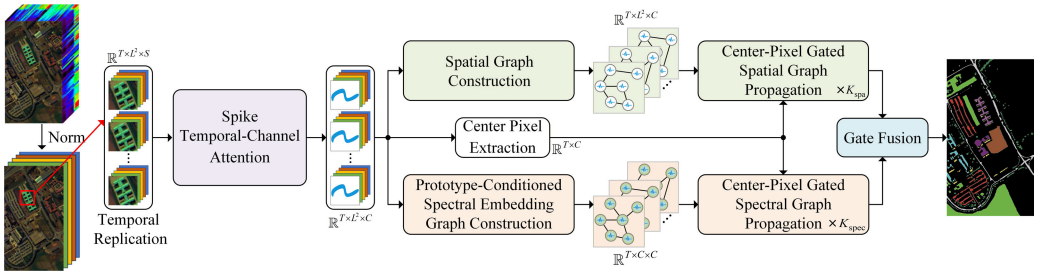


Fig. 1. Overall framework of the proposed CGS^{3}GCN. STCA first recalibrates temporal–channel spike representations, followed by dual-branch spatial and PCSEG reasoning with center-pixel gated propagation. The learned representations are adaptively fused for classification.

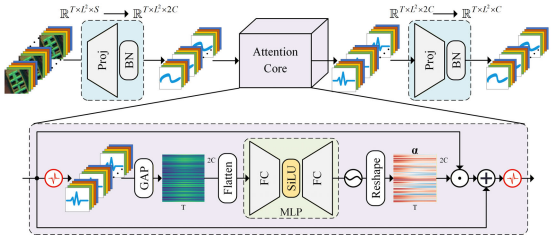


Fig. 2. Illustration of the proposed STCA module. SGS are utilized for CT to adaptively recalibrate temporal–channel feature representations.

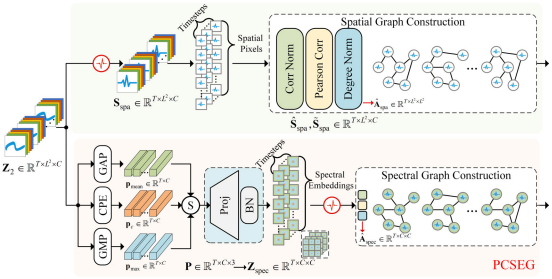


Fig. 3. Illustration of the proposed dual-branch spatial–spectral spike graph construction. The spatial branch captures non-Euclidean contextual relationships via a Pearson-correlation spatial graph, while the spectral branch builds a PCSEG from center-, mean-, and max-prototype-conditioned latent spectral embeddings to model compact spectral dependencies.

Hierarchical graph convolutions are then performed in parallel, and the resulting spatial and spectral representations are adaptively fused via a gated mechanism for final classification. The entire framework is trained end-to-end.

C. Spike Temporal–Channel Attention

To better exploit the temporal information embedded in spike sequences, we introduce an STCA module, as illustrated in Fig. 2. Existing SNN-based HSIC methods mainly utilize spike responses for feature propagation, while the global firing statistics and temporal–channel dependencies encoded in spike sequences remain underexplored. To address this issue, STCA leverages SGS and CT to generate adaptive temporal–channel attention for membrane potential recalibration before graph construction.

Given the temporally replicated input feature $\mathbf{X}^{(T)} \in \mathbb{R}^{T\times L^2\times S}$, where T denotes the number of time steps, S is the spectral dimension, and L^2 represents the number of spatial locations, a linear spectral projection is first applied

$$\mathbf{Z}_1 = \mathrm{BN}\left(\mathbf{X}^{(T)}\mathbf{W}_1\right) \tag{5}$$

where $\mathbf{W}_1 \in \mathbb{R}^{S\times 2C}$ is the learnable projection matrix with C denoting the latent embedding dimension, yielding $\mathbf{Z}_1 \in \mathbb{R}^{T\times L^2\times 2C}$. The projected membrane potentials are then encoded into spike sequences

$$\mathbf{S}_1 = \mathrm{LIF}\left(\mathbf{Z}_1\right) \in \mathbb{R}^{T\times L^2\times 2C}. \tag{6}$$

To obtain SGS, spatial global average pooling is applied to the spike responses

$$\mathbf{R} = \frac{1}{L^2}\sum_{i=1}^{L^2}\mathbf{S}_1\left[:,i,:\right] \in \mathbb{R}^{T\times 2C}. \tag{7}$$

To perform CT, the SGS are vectorized and fed into a bottleneck multilayer perceptron (MLP). The vectorization operator vec$(\cdot)$ flattens the temporal and channel dimensions of $\mathbf{R}$ into a single vector of length $2CT$, and the subsequent fully connected MLP layers capture cross-timestep channel dependencies implicitly through dense mixing of all entries

$$\boldsymbol{\alpha} = \sigma\left(\mathrm{SiLU}\left(\mathrm{vec}\left(\mathbf{R}\right)\mathbf{W}_{\mathrm{down}} + \boldsymbol{b}_{\mathrm{down}}\right)\mathbf{W}_{\mathrm{up}} + \boldsymbol{b}_{\mathrm{up}}\right) \tag{8}$$

where $\boldsymbol{\alpha} \in \mathbb{R}^{2CT}$ denotes the temporal–channel attention vector, $\sigma(\cdot)$ denotes the sigmoid activation function that maps attention scores into the range $(0, 1)$, and SiLU$(\cdot)$ denotes the intermediate activation function defined as $\mathrm{SiLU}(x) = x\sigma(x)$. $\mathbf{W}_{\mathrm{down}} \in \mathbb{R}^{2CT\times(2CT)/r}$ and $\mathbf{W}_{\mathrm{up}} \in \mathbb{R}^{(2CT)/r\times 2CT}$ are the learnable weight matrices of the bottleneck MLP, $r = 4$ denotes the reduction ratio, and $\boldsymbol{b}_{\mathrm{down}}$ and $\boldsymbol{b}_{\mathrm{up}}$ represent learnable bias terms. The obtained attention is reshaped into $\mathbb{R}^{T\times 1\times 2C}$ and broadcast along the spatial dimension.

The membrane potentials are recalibrated in a residual manner

$$\tilde{\mathbf{Z}}_1 = \mathbf{Z}_1 \odot (1 + \boldsymbol{\alpha}) \tag{9}$$

where $\tilde{\mathbf{Z}}_1$ retains the same dimensions as $\mathbf{Z}_1$, followed by spike re-encoding and feature compression

$$\mathbf{Z}_2 = \mathrm{BN}\left(\mathrm{LIF}\left(\tilde{\mathbf{Z}}_1\right)\mathbf{W}_2\right) \tag{10}$$

where $\mathbf{W}_2 \in \mathbb{R}^{2C\times C}$, yielding $\mathbf{Z}_2 \in \mathbb{R}^{T\times L^2\times C}$.

By jointly exploiting SGS and CT, STCA enhances temporal–channel feature discrimination and provides more informative spike representations for subsequent graph reasoning.

D. Dual-Branch Spatial–Spectral Graph Construction

Based on the STCA-enhanced membrane potential representation $\mathbf{Z}_2$, dual-branch spatial and spectral graphs are

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constructed for subsequent graph reasoning, as illustrated in Fig. 3.

1) Spatial Graph: The spatial branch models non-Euclidean spatial correlations among neighboring pixels. Specifically, the membrane potentials are first encoded into spatial spike representations

$$\mathbf{S}_{\text{spa}} = \text{LIF}(\mathbf{Z}_2) \in \mathbb{R}^{T\times L^2\times C}. \tag{11}$$

A Pearson correlation-based strategy is adopted to construct the spatial graph from spike representations. Channelwise centralization is first performed

$$\widehat{\mathbf{S}}_{\text{spa}} = \mathbf{S}_{\text{spa}} - \frac{1}{C}\sum_{i=1}^{C}\mathbf{S}_{\text{spa}}[:,:,i] \tag{12}$$

followed by ℓ_2 normalization

$$\widetilde{\mathbf{S}}_{\text{spa}} = \frac{\widehat{\mathbf{S}}_{\text{spa}}}{\left\|\widehat{\mathbf{S}}_{\text{spa}}\right\|_2} \tag{13}$$

where $\widehat{\mathbf{S}}_{\text{spa}}, \widetilde{\mathbf{S}}_{\text{spa}} \in \mathbb{R}^{T\times L^2\times C}$.

The spatial adjacency matrix is then computed as

$$\mathbf{A}_{\text{spa}} = \text{ReLU}\left(\widetilde{\mathbf{S}}_{\text{spa}}\widetilde{\mathbf{S}}_{\text{spa}}^{\top}\right) \tag{14}$$

where $\mathbf{A}_{\text{spa}} \in \mathbb{R}^{T\times L^2\times L^2}$. Subsequently, symmetric degree normalization is performed

$$\widehat{\mathbf{A}}_{\text{spa}} = \mathbf{D}_{\text{spa}}^{-\frac{1}{2}}\mathbf{A}_{\text{spa}}\mathbf{D}_{\text{spa}}^{-\frac{1}{2}} \tag{15}$$

where $\mathbf{D}_{\text{spa}} \in \mathbb{R}^{T\times L^2\times L^2}$ denotes the degree matrix of $\mathbf{A}_{\text{spa}}$.

The normalized adjacency matrix $\widehat{\mathbf{A}}_{\text{spa}} \in \mathbb{R}^{T\times L^2\times L^2}$ is subsequently used for spatial spike graph propagation.

2) Prototype-Conditioned SEG: Conventional spectral graphs construct node representations directly from channelwise spatial responses, which describe each spectral channel in isolation and therefore lack explicit semantic grounding. To address this limitation, we propose a PCSEG, which conditions spectral node representations on complementary prototypes extracted from the same hyperspectral patch, rather than deriving them from raw channelwise features alone. Specifically, a center prototype captures target-aware semantics directly associated with the classification label, a mean prototype summarizes the overall contextual spectral distribution, and a max prototype emphasizes dominant salient responses. By referencing these semantically anchored prototypes, the resulting node descriptors encode how each spectral embedding channel relates to informative statistical perspectives of the patch, producing compact and discriminative representations for spectral graph reasoning.

Specifically, three complementary prototypes are extracted to characterize spectral distributions from different perspectives. The center prototype preserves the semantic characteristics of the target pixel, which is directly associated with the classification label. The mean prototype summarizes the overall contextual distribution within the local neighborhood, while the max prototype emphasizes dominant spectral responses and salient structures.

The center prototype is defined as

$$\mathbf{p}_c = \mathbf{Z}_2[:, l_c, :] \in \mathbb{R}^{T\times C} \tag{16}$$

where l_c denotes the center spatial position of the local patch.

Similarly, the mean and max prototypes are, respectively, computed as

$$\mathbf{p}_{\text{mean}} = \frac{1}{L^2}\sum_{i=1}^{L^2}\mathbf{Z}_2[:,i,:] \in \mathbb{R}^{T\times C} \tag{17}$$

$$\mathbf{p}_{\text{max}} = \max_{i=1}^{L^2}\mathbf{Z}_2[:,i,:] \in \mathbb{R}^{T\times C}. \tag{18}$$

The three prototypes are stacked along a new dimension to form

$$\mathbf{P} = \left[\mathbf{p}_c, \mathbf{p}_{\text{mean}}, \mathbf{p}_{\text{max}}\right] \in \mathbb{R}^{T\times C\times 3}. \tag{19}$$

The complementary prototypes are then projected into a latent embedding space to generate compact spectral node representations

$$\mathbf{Z}_{\text{spec}} = \text{BN}\left(\mathbf{P}\mathbf{W}_{\text{proj}}\right) \tag{20}$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{3\times C}$, yielding $\mathbf{Z}_{\text{spec}} \in \mathbb{R}^{T\times C\times C}$.

Intuitively, the projection matrix $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{3\times C}$ performs a learned linear combination of the three complementary prototypes, mapping them from the 3-D prototype space into the C-dimensional spectral embedding space. Each resulting spectral node embedding thus encodes a weighted mixture of target-aware semantics (center), contextual distribution (mean), and salient spectral responses (max), with the mixing coefficients adaptively learned to capture discriminative spectral dependencies.

Following common spectral graph modeling strategies, each projected spectral embedding channel is treated as a graph node. Different from conventional spectral graphs that directly derive node representations from channelwise spatial responses, the proposed node features are generated from prototype-conditioned latent embeddings.

The projected spectral membrane potentials are encoded into spectral spike representations

$$\mathbf{S}_{\text{spec}} = \text{LIF}\left(\mathbf{Z}_{\text{spec}}\right). \tag{21}$$

Following the graph construction strategy in the spatial branch, the normalized spectral adjacency matrix $\widehat{\mathbf{A}}_{\text{spec}} \in \mathbb{R}^{T\times C\times C}$ is constructed based on the spectral embedding spike representations $\mathbf{S}_{\text{spec}}$, enabling spectral dependency reasoning in the latent embedding space.

Although the spatial and spectral graphs encode different semantic relationships, both are constructed in the same C-dimensional latent embedding space and describe pairwise feature-similarity structures among nodes—spatial positions in one case and spectral embedding channels in the other. Adopting the same Pearson-correlation-based strategy with channelwise centralization, ℓ_2 normalization, and symmetric degree normalization is therefore well motivated: it provides a consistent, scale-invariant measure of node affinity for both branches, avoids introducing branch-specific hyperparameters, and keeps the two adjacency matrices at a comparable normalization scale, which facilitates the subsequent adaptive fusion of the spatial and spectral representations.

E. Center-Pixel Gated Spike Graph Propagation

Although graph propagation effectively aggregates contextual information, repeated message passing may progressively dilute the class-defining semantic signal associated with the classification target. In the spatial branch, heterogeneous

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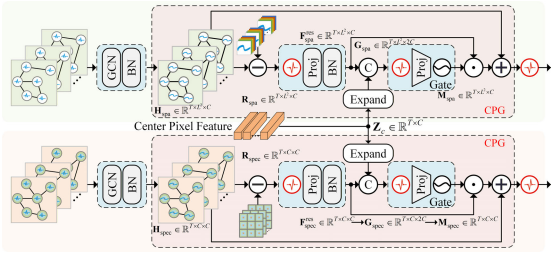


Fig. 4. Illustration of spike graph propagation with the proposed CPG. Center-pixel semantics are utilized to adaptively refine graph representations and preserve target-oriented discriminative information during propagation.

neighboring pixels can introduce interfering features; in the spectral branch, interchannel mixing can blend redundant spectral responses into target-related representations. Since the center pixel's spectral–spatial signature directly determines the classification label in patch-based HSIC, any contamination of this signature during graph propagation undermines the discriminability of the learned representation. To address this issue, we introduce a CPG, which uses the center-pixel representation as adaptive semantic guidance to refine graph propagation in both spatial and spectral branches, as shown in Fig. 4.

1) Spatial–Spectral Spike Graph Propagation: Based on the constructed adaptive spatial–spectral spike graphs, spike graph propagation is performed to aggregate contextual node dependencies.

For the spatial branch, spike graph propagation is formulated as

$$\mathbf{H}_{\text{spa}} = \text{BN}\left(\widehat{\mathbf{A}}_{\text{spa}}\mathbf{S}_{\text{spa}}\mathbf{W}_{\text{spa}}\right) \tag{22}$$

where $\mathbf{W}_{\text{spa}} \in \mathbb{R}^{C\times C}$ denotes the learnable graph projection matrix and $\mathbf{H}_{\text{spa}} \in \mathbb{R}^{T\times L^2\times C}$ represents the propagated spatial membrane potentials.

Similarly, spectral spike graph propagation is performed as

$$\mathbf{H}_{\text{spec}} = \text{BN}\left(\widehat{\mathbf{A}}_{\text{spec}}\mathbf{S}_{\text{spec}}\mathbf{W}_{\text{spec}}\right) \tag{23}$$

where $\mathbf{W}_{\text{spec}} \in \mathbb{R}^{C\times C}$, yielding $\mathbf{H}_{\text{spec}} \in \mathbb{R}^{T\times C\times C}$.

2) Center-Pixel Gate: The CPG refines graph representations according to the semantic relevance between graph nodes and the center pixel.

For the spatial branch, a residual representation is first computed between the original membrane potentials and graph-propagated features

$$\mathbf{R}_{\text{spa}} = \mathbf{Z}_2 - \mathbf{H}_{\text{spa}} \in \mathbb{R}^{T\times L^2\times C} \tag{24}$$

which highlights feature components that are weakened during neighborhood aggregation. Specifically, graph propagation aggregates features from neighboring nodes, which homogenizes the representation and can attenuate the class-defining signature of the center pixel. Subtracting the propagated features from the original membrane potentials therefore isolates the discriminative components that aggregation has weakened, providing a targeted cue for the gate to selectively restore. This design enables the CPG to reinforce the target-oriented information lost during propagation rather than injecting the full original features indiscriminately. The residual features are then refined as

$$\mathbf{F}^{\text{res}}_{\text{spa}} = \text{BN}\left(\text{LIF}\left(\mathbf{R}_{\text{spa}}\right)\mathbf{W}_{\text{res}}\right) \tag{25}$$

where $\mathbf{W}_{\text{res}} \in \mathbb{R}^{C\times C}$ is a learnable projection matrix and $\mathbf{F}^{\text{res}}_{\text{spa}} \in \mathbb{R}^{T\times L^2\times C}$.

The LIF encoding of these residual cues remains effective for two reasons. First, the firing threshold acts as a sparsification mechanism that retains only the residual components exceeding the threshold, thereby suppressing weak, noise-like differences introduced by neighborhood aggregation. Second, encoding the residual into spikes keeps it in the same spiking domain as the rest of the framework and preserves temporal consistency across the T timesteps, allowing the residual information to be integrated with the propagated spike representations in subsequent stages.

The center-pixel representation is extracted as

$$\mathbf{Z}_c = \mathbf{Z}_2\left[:, l_c, :\right] \in \mathbb{R}^{T\times C} \tag{26}$$

where l_c denotes the center spatial position of the local patch. The center representation is spatially broadcast to match the graph-node dimension, yielding $\widetilde{\mathbf{Z}}_c \in \mathbb{R}^{T\times L^2\times C}$, which is concatenated with the residual features

$$\mathbf{G}_{\text{spa}} = \left[\mathbf{F}^{\text{res}}_{\text{spa}}, \widetilde{\mathbf{Z}}_c\right] \in \mathbb{R}^{T\times L^2\times 2C}. \tag{27}$$

A center-guided gating mask is generated as

$$\mathbf{M}_{\text{spa}} = \sigma\left(\text{LIF}\left(\mathbf{G}_{\text{spa}}\right)\mathbf{W}_g\right) \in \mathbb{R}^{T\times L^2\times C} \tag{28}$$

where $\mathbf{W}_g \in \mathbb{R}^{2C\times C}$.

The gated spatial representation is obtained by

$$\widetilde{\mathbf{H}}_{\text{spa}} = \mathbf{H}_{\text{spa}} + \mathbf{M}_{\text{spa}} \odot \mathbf{F}^{\text{res}}_{\text{spa}}. \tag{29}$$

The refined membrane potentials are subsequently encoded into spike representations for the next propagation stage

$$\mathbf{S}^{\text{out}}_{\text{spa}} = \text{LIF}\left(\widetilde{\mathbf{H}}_{\text{spa}}\right). \tag{30}$$

Similarly, the CPG is applied to the spectral branch, where the center prototype $\mathbf{p}_c$ serves as semantic guidance for adaptive spectral graph refinement.

For intermediate graph layers, the output spike representations are used as inputs to subsequent propagation stages. For the final layer, the LIF operation is omitted to avoid information loss before classification. The continuous membrane potentials $\widetilde{\mathbf{H}}_{\text{spa}}$ and $\widetilde{\mathbf{H}}_{\text{spec}}$ are therefore directly adopted as the final outputs of the spatial and spectral branches.

By using the center-pixel representation as adaptive semantic guidance, CPG preserves target-oriented semantics during graph propagation and improves both spatial and spectral graph reasoning.

F. Dynamic Spatial–Spectral Fusion

The spatial and spectral branches provide complementary representations. Therefore, a dynamic gating mechanism is employed for adaptive fusion, as shown in Fig. 5.

Given the obtained branch outputs $\mathbf{H}^{\text{out}}_{\text{spa}} \in \mathbb{R}^{T\times L^2\times C}$ and $\mathbf{H}^{\text{out}}_{\text{spec}} \in \mathbb{R}^{T\times C\times C}$, the spatial features are first globally pooled along spatial graph nodes

$$\mathbf{f}_{\text{spa}} = \frac{1}{L^2}\sum_{i=1}^{L^2}\mathbf{H}^{\text{out}}_{\text{spa}}\left[:, i, :\right] \in \mathbb{R}^{T\times C}. \tag{31}$$

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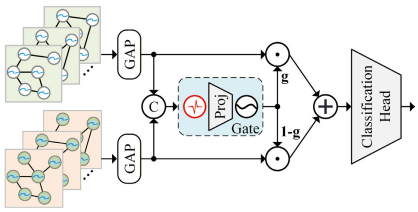


Fig. 5. Illustration of the gated spatial–spectral fusion mechanism. Spatial and spectral representations are adaptively integrated through learnable spike-driven fusion coefficients.

Similarly, the spectral features are globally pooled along the feature dimension

$$\mathbf{f}_{\text{spec}} = \frac{1}{C}\sum_{i=1}^{C}\mathbf{H}_{\text{spec}}^{\text{out}}[:,:,i] \in \mathbb{R}^{T\times C}. \tag{32}$$

Fusion coefficients are computed as

$$\mathbf{g} = \sigma\left(\text{LIF}\left(\left[\mathbf{f}_{\text{spa}}, \mathbf{f}_{\text{spec}}\right]\right)\mathbf{W}_f\right) \tag{33}$$

where $\mathbf{W}_f \in \mathbb{R}^{2C\times C}$ is the gating projection matrix and $\mathbf{g} \in \mathbb{R}^{T\times C}$ denotes the fusion coefficients.

The fused spatial–spectral representation is obtained as

$$\mathbf{f} = \mathbf{g}\odot\mathbf{f}_{\text{spa}} + (1-\mathbf{g})\odot\mathbf{f}_{\text{spec}}. \tag{34}$$

Finally, the fused representation is fed into a linear classifier

$$\mathbf{O}_t = \text{FC}(\mathbf{f}[t,:]) \tag{35}$$

where $\mathbf{O}_t \in \mathbb{R}^{N_c}$ denotes the classification logits at the tth time step and N_c is the number of land-cover categories.

G. Training Objective

Following the temporal spiking learning paradigm, classification supervision is imposed at each simulation time step. Specifically, given the ground-truth label y, the temporal classification loss at the tth time step is formulated as $\mathcal{L}_{CE}(\mathbf{O}_t, y)$.

The total training objective is obtained by averaging the temporal losses over all simulation steps

$$\mathcal{L}_{\text{total}} = \frac{1}{T}\sum_{t=1}^{T}\mathcal{L}_{\text{CE}}(\mathbf{O}_t, y). \tag{36}$$

During inference, the final prediction is obtained by averaging temporal classification logits across all time steps

$$\bar{\mathbf{O}} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{O}_t. \tag{37}$$

The predicted category is then determined by

$$\hat{y} = \arg\max_c \bar{\mathbf{O}}_c. \tag{38}$$

IV. Experiments

In this section, extensive experiments are conducted on four widely used HSIC benchmarks to comprehensively evaluate the effectiveness of the proposed CGS^{3}GCN framework.

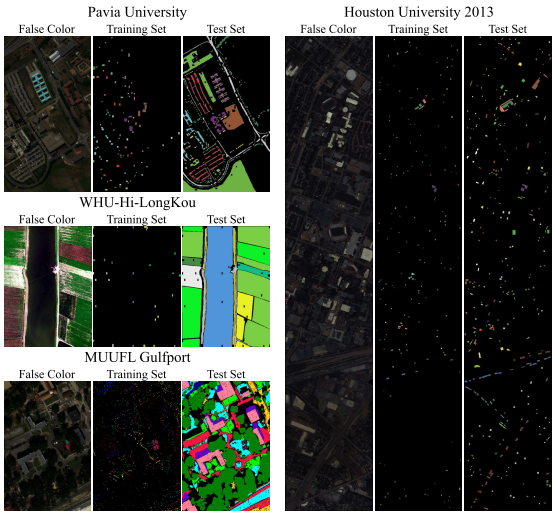


Fig. 6. False-color images and visualizations of the spatially disjoint training/test splits for the four datasets.

A. Dataset Description

Experiments are conducted on four widely used HSIC benchmarks: 1) Pavia University (PU) [41], which contains 610 × 340 pixels with 103 spectral bands and nine land-cover classes; 2) Houston University 2013 (HU) [42], consisting of 349 × 1905 pixels with 144 spectral bands and 15 classes; 3) WHU-Hi-LongKou (WHLK) [43], containing 550 × 400 pixels with 270 spectral bands and nine classes; and 4) MUUFL Gulfport [44], [45], which consists of 325 × 220 pixels with 64 spectral bands and 11 classes.

To avoid over-optimistic evaluation caused by spatial correlation leakage in random sampling [46], spatially disjoint train-test partitions are adopted for all datasets. The partition settings are consistent with those used in [21], [39], and [47]. Detailed sample distributions and visual scenes are provided in Table I and Fig. 6.

B. Experimental Setup

Evaluation Metrics: Overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ) are adopted for quantitative evaluation, where higher values indicate better classification performance.

Compared Methods: The proposed CGS^{3}GCN is compared with nine SOTA baselines from four categories.

1) *Transformer-Based Methods:* MsET [48], MSST [49], and DSFormer [50].
2) *Frequency-Domain Methods:* MSFI-CNet [51] and SFMamba [52].
3) *Graph-Based Methods:* SSGRAM [30] and MS2FN [21].
4) *SNN-Based Methods:* AFSA [37] and FRM-SNN [35].

For fairness, all competing methods are implemented using the recommended settings reported in their original papers.

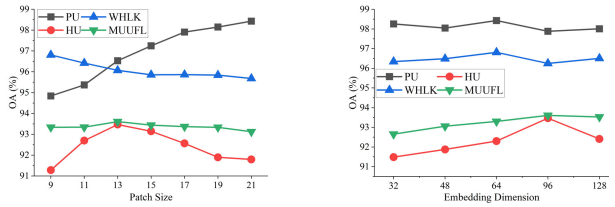
Implementation Details: Experiments are conducted on an Ubuntu server with an NVIDIA RTX 3090 GPU (24 GB) using PyTorch 2.0.1. Spike-driven computation is implemented via SpikingJelly [40]. The network is trained for 100 epochs using Adam with a batch size of 64. The learning rate is initialized

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TABLE I

LAND-COVER CLASSES AND SAMPLE NUMBERS FOR THE SPATIALLY DISJOINT TRAINING AND TEST SPLITS ACROSS THE FOUR DATASETS

NO.	Pavia University			Houston University 2013			WHU-Hi-LongKou			MUUFL Gulfport		
	Class	Train	Test	Class	Train	Test	Class	Train	Test	Class	Train	Test
1	Asphalt	548	6304	Healthy Grass	198	1053	Corn	200	34311	Trees	150	23096
2	Meadows	540	18146	Stressed Grass	190	1064	Cotton	200	8174	Mostly Grass	150	4120
3	Gravel	392	1815	Synthetic Grass	192	505	Sesame	200	2831	Mixed Ground Surface	150	6732
4	Trees	524	2912	Tree	188	1056	Broad-leaf Soybean	200	63012	Dirt and Sand	150	1676
5	Metal Sheets	265	1113	Soil	186	1056	Narrow-leaf Soybean	200	3951	Road	150	6537
6	Bare Soil	532	4572	Water	182	143	Rice	200	11654	Water	150	316
7	Bitumen	375	981	Residential	196	1072	Water	200	66856	Building Shadow	150	2083
8	Bricks	514	3364	Commercial	191	1053	Roads and Houses	200	6924	Building	150	6090
9	Shadows	231	795	Road	193	1059	Mixed Weed	200	5029	Sidewalk	150	1235
10				Highway	191	1036				Yellow Curb	150	33
11				Railway	181	1054				Cloth Panels	150	119
12				Parking Lot1	192	1041						
13				Parking Lot2	184	285						
14				Tennis Court	181	247						
15				Running Track	187	473						
	Total	3921	40002	Total	2832	12197	Total	1800	202742	Total	1650	52037



(a) (b)

Fig. 7. Sensitivity analysis of (a) spatial patch size and (b) embedding dimension C.

TABLE II

ROBUSTNESS OF STCA UNDER DIFFERENT TIMESTEPS SETTINGS [OA (%)]

T	PU		HU		WHLK		MUUFL	
	w/o	w/	w/o	w/	w/o	w/	w/o	w/
1	95.98	97.90	91.36	92.21	94.76	96.15	92.94	93.18
2	97.52	98.20	92.06	93.11	95.65	96.37	93.31	93.50
3	97.27	**98.43**	92.03	**93.47**	95.83	96.52	93.23	**93.61**
4	97.14	98.18	91.97	92.22	96.06	**96.81**	93.30	93.49
5	97.43	97.49	92.24	92.46	95.68	96.78	93.27	93.47

Note: w/ and w/o denote with and without STCA, respectively.

to 5×10^{-4} and decayed to 1×10^{-6} via cosine annealing. During inference, the model with the minimum training loss is used for evaluation. To ensure reliable evaluation, all experiments are independently repeated five times with different random seeds, and the average results are reported.

C. Parameter Analysis

This section evaluates the influence of several key hyperparameters on CGS^{3}GCN, including the spatial patch size, embedding dimension, graph propagation depths, timesteps, and neuronal dynamics.

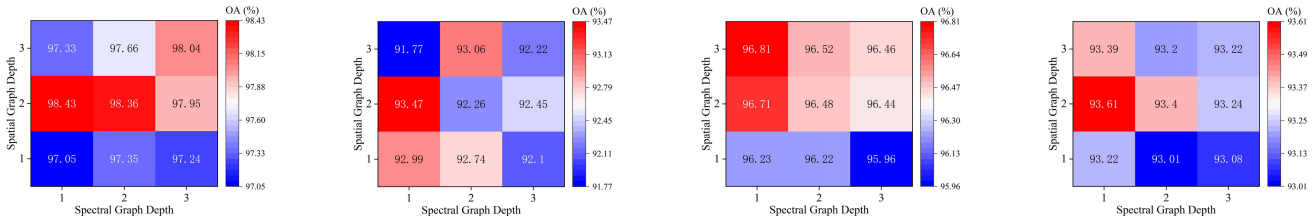
As shown in Fig. 7(a), the optimal patch size varies across datasets. PU benefits from a larger receptive field due to broader spatial structures, whereas HU, WHLK, and MUUFL achieve better performance with moderate patch sizes. Excessively large neighborhoods introduce heterogeneous interference and weaken target-oriented semantic representations, highlighting the importance of preserving center-pixel discriminability during graph propagation.

The embedding dimension C controls both feature representation capacity and spectral embedding graph (SEG) scale. As illustrated in Fig. 7(b), the best performance is achieved at $C = 64$ for PU and WHLK, and $C = 96$ for HU and MUUFL. Smaller embedding dimensions restrict representation capacity, whereas excessively large dimensions introduce redundant spectral information and increase optimization difficulty. The results indicate that compact spectral embeddings provide a favorable tradeoff between representation capacity and redundancy reduction, validating the design motivation of PCSEG.

As shown in Fig. 8, the spectral branch consistently prefers shallow propagation ($K_{\text{spec}} = 1$). This result suggests that PCSEG already captures discriminative spectral dependencies through prototype-conditioned node representations. From the perspective of graph over-smoothing, the spectral graph is relatively compact and semantically rich due to the three prototype conditions; hence, a single round of message passing among all nodes is sufficient to aggregate cross-channel spectral dependencies, while further increasing the propagation depth only over-smooths the spectral nodes without yielding better performance. In contrast, the spatial branch benefits from deeper propagation ($K_{\text{spa}} = 2$ or 3), which enlarges the receptive field and facilitates long-range spatial interaction modeling.

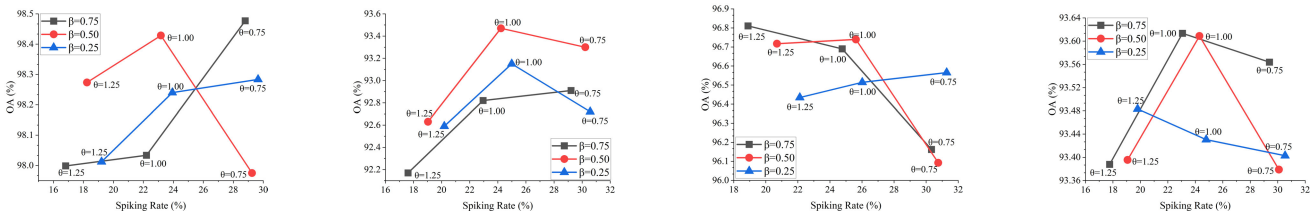
As shown in Table II, the optimal setting is $T = 3$ for PU, HU, and MUUFL, and $T = 4$ for WHLK. Further increasing the number of timesteps yields marginal gains, indicating that sufficient temporal dynamics can be captured with a small number of propagation steps. Moreover, STCA consistently improves performance across all timesteps settings, demonstrating its robustness to different temporal durations.

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(a) (b) (c) (d)

Fig. 8. Sensitivity analysis of spatial (K_{spa}) and spectral (K_{spec}) graph propagation depths. (a) PU. (b) HU. (c) WHLK. (d) MUUFL.



(a) (b) (c) (d)

Fig. 9. Sensitivity analysis of LIF neuronal parameters (β and θ). (a) PU. (b) HU. (c) WHLK. (d) MUUFL.

Finally, the leakage factor β controls temporal information retention, while the firing threshold θ determines spike activation sparsity. As shown in Fig. 9, excessively small values lead to insufficient temporal accumulation, whereas overly large values suppress spike generation and reduce representation diversity. The optimal settings achieve an effective tradeoff between temporal information retention and spike sparsity, resulting in more discriminative spiking representations.

Overall, the results validate the effectiveness of the proposed PCSEG, STCA, and CPG designs. The optimal parameter settings adopted in subsequent experiments are summarized as follows. The patch size is set to 21, 13, 9, and 13 for PU, HU, WHLK, and MUUFL, respectively. The embedding dimension is set to 64 for PU and WHLK and 96 for HU and MUUFL. The graph propagation depths are configured as $(K_{spa}, K_{spec}) = (2, 1)$ for PU, HU, and MUUFL, and $(3, 1)$ for WHLK. The timesteps are set to 3 for PU, HU, and MUUFL and 4 for WHLK. Moreover, the optimal tradeoff between temporal information retention and spike sparsity is achieved with $(\beta, \theta) = (0.50, 1.00)$ for PU and HU, $(0.75, 1.25)$ for WHLK, and $(0.75, 1.00)$ for MUUFL.

D. Ablation Study

To validate the effectiveness of the proposed CGS^{3}GCN framework, extensive ablation studies are conducted on the four datasets, including analyses of algorithmic components, temporal–channel attention design, spectral graph representations, prototype configurations, and center-pixel-guided feature refinement.

1) Effectiveness of Individual Algorithmic Components: As reported in Table III, each proposed component contributes positively to the overall performance of CGS^{3}GCN.

Introducing STCA (Model 2) consistently improves the baseline (Model 1) on all datasets, demonstrating that exploiting SGS and cross-timestep channel dependencies provides more discriminative temporal–channel representations.

TABLE III
ABLATION STUDY OF DIFFERENT FRAMEWORK COMPONENTS [OA (%)]

ID	STCA	Spa.	Spe.	CPG	PU	HU	WHLK	MUUFL
1					85.02	86.00	94.50	82.99
2	✓				90.28	88.23	95.53	84.63
3	✓	✓			92.55	89.12	95.92	86.06
4	✓		✓		95.61	91.58	95.93	92.46
5	✓	✓	✓		97.94	91.99	96.40	93.20
6	✓	✓		✓	97.25	91.73	96.50	93.47
7	✓		✓	✓	96.54	92.18	96.13	92.72
8	✓	✓	✓	✓	**98.43**	**93.47**	**96.81**	**93.61**

When graph reasoning is introduced, both the spatial branch (Model 3) and the spectral branch (Model 4) further improve performance. Notably, the spectral branch yields larger gains on most datasets, indicating that PCSEG effectively captures discriminative spectral dependencies through prototype-conditioned node representations and alleviates the limitations of conventional spectral graph representations.

Combining the spatial and spectral branches (Model 5) leads to additional improvements, confirming that the two branches provide complementary spatial-contextual and spectral dependency information.

Although the spectral branch alone is already competitive, the spatial branch contributes most in two scenarios: when adjacent classes have similar spectral signatures, where spatial context disambiguates the center pixel; and when scenes contain strong spatial structures, where neighborhood aggregation smooths the prediction and suppresses isolated errors. This is reflected in the ablation results—the largest gain from adding the spatial branch is on PU (+2.33%, from 95.61% to 97.94%), which exhibits stronger spatial structure, while the gains are smaller on HU (+0.41%) and WHLK (+0.47%), where spectral dependencies already discriminate well. The spatial

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TABLE IV
COMPONENTWISE ANALYSIS OF THE PROPOSED STCA MODULE [OA (%)]

Setting	PU	HU	WHLK	MUUFL
None	97.27	92.03	96.06	93.23
SE	98.20	92.26	96.49	93.50
CT-SE	97.82	93.39	96.19	93.47
SGS-SE	98.39	92.29	96.78	93.50
STCA	**98.43**	**93.47**	**96.81**	**93.61**

branch thus remains valuable because it provides contextual information that global spectral statistics cannot capture.

Furthermore, introducing CPG into either the spatial branch (Model 6) or spectral branch (Model 7) consistently enhances classification accuracy compared with their corresponding counterparts (Models 3 and 4). This demonstrates that preserving target-oriented semantics during graph propagation effectively mitigates semantic dilution of the center pixel.

Finally, integrating STCA, PCSEG, and CPG into a unified spatial–spectral spiking graph framework (Model 8) achieves the best performance on all datasets, validating the effectiveness and complementarity of the proposed design.

2) Componentwise Analysis of STCA: To further validate the design of STCA, we separately evaluate the contributions of SGS and CT. The results are reported in Table IV.

As shown in Table IV, all attention variants outperform the baseline without attention, demonstrating the effectiveness of temporal–channel feature recalibration for spiking hyperspectral representation learning. Compared with the conventional SE module [53], SGS-SE consistently achieves better performance on most datasets, indicating that SGS provides more discriminative cues than conventional feature statistics for attention generation.

CT-SE introduces cross-timestep channel interactions and achieves noticeable improvements on HU and MUUFL, but its effectiveness is less consistent when used alone. This suggests that modeling temporal dependencies without informative spike statistics is insufficient for robust temporal–channel recalibration.

The complete STCA consistently achieves the best performance across all datasets. These results demonstrate that SGS and CT are complementary: SGS captures informative temporal firing statistics, while CT further exploits cross-timestep channel dependencies for attention refinement. Their combination enables more effective utilization of spike temporal dynamics and leads to superior classification performance.

3) Analysis of Spectral Representations: To investigate the effectiveness of different spectral representations, we compare four spectral graph construction strategies. Specifically, raw spectral graph (RSG) directly treats the original spectral bands as graph nodes, while SEG constructs the graph in the latent spectral embedding space. Furthermore, prototype-conditioned representation (PCR) introduces prototype representations for node construction.

As reported in Table V, the effectiveness of spectral graph learning is highly dependent on the quality of node representations. Although SEG constructs graphs in a compact latent embedding space, it does not consistently outperform

TABLE V
ANALYSIS OF DIFFERENT SPECTRAL REPRESENTATIONS [OA (%)]

RSG	SEG	PCR	PU	HU	WHLK	MUUFL
✓			97.00	91.98	96.11	92.53
	✓		95.87	92.24	95.81	91.80
✓		✓	98.10	91.99	96.37	93.48
	✓	✓	**98.43**	**93.47**	**96.81**	**93.61**

TABLE VI
ANALYSIS OF DIFFERENT PROTOTYPE CONFIGURATIONS [OA (%)]

Center	Mean	Max	PU	HU	WHLK	MUUFL
✓			80.96	83.83	90.11	81.98
	✓		91.38	88.38	95.06	84.01
		✓	82.64	83.09	93.14	76.02
✓	✓		94.88	89.34	95.56	90.91
✓		✓	94.44	90.37	94.70	90.17
	✓	✓	91.62	89.80	95.27	84.40
✓	✓	✓	**95.28**	**91.36**	**95.69**	**91.95**

RSG, indicating that reducing spectral redundancy alone is insufficient for discriminative spectral dependency modeling.

In contrast, introducing PCR consistently improves the performance of both RSG and SEG. This result suggests that prototype-conditioned node representations provide more discriminative semantic cues than directly using channelwise spatial responses. By incorporating complementary prototype information, PCR enables more informative spectral node construction and facilitates more effective spectral graph reasoning.

The combination of SEG and PCR achieves the best performance on all datasets, demonstrating that compact spectral embeddings and prototype-conditioned node representations are both important for effective spectral graph learning. These results validate the design motivation of the proposed PCSEG.

To further investigate the effectiveness of different prototype configurations, we evaluate various prototype combinations within the proposed PCR. Since the proposed framework adopts a dual-branch architecture with adaptive fusion, the contribution of individual prototypes may be partially compensated by the spatial branch. Therefore, only the spectral graph branch is retained in this experiment, while STCA and CPG are disabled to isolate the effect of different prototype configurations.

As reported in Table VI, different prototype configurations exhibit distinct discriminative capabilities. Among the single-prototype settings, the Mean prototype consistently achieves the best performance, indicating that contextual statistical information provides a stable basis for spectral node representation learning. The Center prototype preserves target-oriented semantics, whereas the Max prototype emphasizes salient spectral responses.

Combining multiple prototypes generally yields better performance than using any single prototype alone, demonstrating that the three prototypes provide complementary semantic information. Specifically, the Center, Mean, and Max prototypes, respectively, encode target-oriented semantics, contextual statistics, and salient responses. Integrating

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TABLE VII
ANALYSIS OF DIFFERENT GATE SOURCES [OA (%)]

Gate Source	PU	HU	WHLK	MUUFL
No-Gate	97.57	93.06	96.54	93.54
Max-Gate	98.13	92.50	96.69	93.52
Mean-Gate	98.36	93.06	96.50	93.55
Center-Gate	**98.43**	**93.47**	**96.81**	**93.61**

these complementary cues produces more discriminative prototype-conditioned node representations for spectral graph construction.

Consequently, the combination of all three prototypes achieves the best performance across all datasets, demonstrating that integrating target-oriented semantics, contextual statistics, and salient responses yields more discriminative spectral node representations. This result further validates the design of PCSEG.

4) Analysis of Different Gate Sources: To validate the effectiveness of center-pixel semantic guidance, we compare different prototype sources for gate generation. Specifically, No-Gate removes the adaptive gating mechanism and directly adds the enhanced residual features to the graph representations without semantic-guided modulation. Mean-Gate and Max-Gate use the mean and max prototypes as semantic guidance to generate gating coefficients, respectively. Center-Gate corresponds to the proposed CPG, where the center-pixel representation is used to guide graph feature refinement. This comparison evaluates whether target-oriented semantics provide more effective guidance than contextual statistical prototypes.

As reported in Table VII, all guided gating strategies generally outperform No-Gate, indicating that semantic-guided modulation is more effective than directly injecting residual information into graph representations. This result validates the necessity of adaptive semantic guidance during graph propagation.

Among different guidance sources, Mean-Gate and Max-Gate provide contextual statistical and salient-response cues, respectively, yielding moderate performance improvements. However, their guidance signals are not directly aligned with the classification target. In contrast, the center pixel is directly associated with the ground-truth label and therefore provides the most task-relevant semantic guidance. Consequently, Center-Gate consistently achieves the best performance across all datasets, demonstrating that preserving target-oriented semantics during graph propagation is crucial for discriminative spatial–spectral graph reasoning. These results validate the design motivation of the proposed CPG.

E. Quantitative Evaluation of Classification Performance

Tables VIII–XI report the classwise accuracies, OA, AA, and κ of all compared methods on the four datasets. Overall, the proposed CGS^{3}GCN consistently achieves the best OA, AA, and κ, demonstrating robust spatial–spectral representation learning across diverse hyperspectral scenes.

Transformer-based methods (e.g., MsET, MSST, and DSFormer) benefit from global dependency modeling but remain less effective in handling irregular structures and fragmented boundaries. Consequently, their performance degrades on challenging categories such as bare soil (Class 6) in PU and highway (Class 10) in HU.

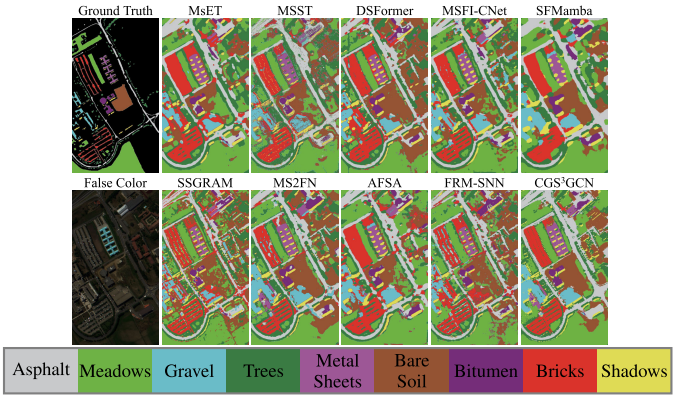


Fig. 10. Visual comparison of classification maps on the PU dataset.

Frequency-domain methods (e.g., MSFI-CNet and SFMamba) enhance feature representations from the spectral-frequency perspective and achieve competitive results on several categories. However, they do not explicitly model long-range non-Euclidean spatial–spectral dependencies, which limits their discriminability in heterogeneous scenes. For example, MSFI-CNet achieves only 61.66% on narrow-leaf soybean (Class 5) in WHLK, while SFMamba obtains 78.54% on residential (Class 7) in HU.

Graph-based methods such as SSGRAM and MS2FN explicitly model contextual relationships and achieve strong performance. Nevertheless, repeated graph aggregation may introduce over-smoothing, while insufficient spectral dependency modeling can limit discriminability on difficult categories, such as highway (Class 10) in HU.

Conventional SNN-based methods (AFSA and FRM-SNN) generally lag behind SOTA ANN-based approaches due to the limited discriminative capability of highly discretized spike representations. For instance, AFSA drops to 35.11% on roads and houses (Class 8) in WHLK, while FRM-SNN achieves only 61.16% on gravel (Class 3) in PU.

In contrast, CGS^{3}GCN jointly integrates temporal–channel spike recalibration, prototype-conditioned spectral graph learning, and center-pixel-guided graph reasoning within a unified spiking framework. By simultaneously exploiting temporal dynamics, spectral dependencies, and target-oriented semantics, the proposed method achieves consistently superior performance across diverse categories and datasets.

F. Visual Comparison of Classification Maps

To qualitatively evaluate classification performance, Figs. 10–13 present the classification maps on the four datasets. Since ground-truth annotations in hyperspectral datasets may contain coarse boundary labeling, reliable classification maps should remain consistent with the actual spatial structures observed in the false-color images.

Overall, most baseline methods exhibit varying degrees of boundary distortion, spatial noise, or category confusion. Transformer-based methods (e.g., MsET, MSST, and DSFormer) often produce fragmented predictions in

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TABLE VIII
CLASSIFICATION PERFORMANCE ON THE PU DATASET

Class NO.	MsET	MSST	DSFormer	MSFI-CNet	SFMamba	SSGRAM	MS2FN	AFSA	FRM-SNN	CGS^{3}GCN
1	90.78	91.53	88.25	93.53	91.32	86.29	91.59	90.37	86.96	**97.25**
2	98.19	95.27	92.95	95.23	98.76	94.58	98.40	97.51	93.60	**98.81**
3	74.55	85.51	80.39	73.44	90.69	75.65	**98.57**	81.54	61.16	95.76
4	96.26	93.65	96.15	94.44	85.44	**98.87**	98.63	79.70	94.09	98.48
5	99.55	99.46	**99.91**	99.19	95.33	98.92	99.46	98.20	99.82	99.66
6	63.15	75.77	**99.28**	80.36	97.86	78.98	95.98	79.44	93.44	98.59
7	87.26	91.13	97.35	91.54	98.57	96.84	94.70	91.23	94.39	**99.76**
8	98.69	96.88	95.39	96.70	97.44	**98.93**	96.61	98.87	98.57	98.73
9	95.60	84.40	96.98	95.22	83.02	96.98	98.74	92.20	95.97	**99.42**
OA (%)	91.57	91.82	93.18	92.36	95.63	91.54	96.87	92.17	91.76	**98.43**
AA (%)	89.34	90.40	94.07	91.07	93.16	91.78	96.96	89.90	90.89	**98.49**
$\kappa \times 100$	88.43	88.94	90.94	89.69	94.10	88.59	95.79	89.35	88.98	**97.88**

TABLE IX
CLASSIFICATION PERFORMANCE ON THE HU 2013 DATASET

Class NO.	MsET	MSST	DSFormer	MSFI-CNet	SFMamba	SSGRAM	MS2FN	AFSA	FRM-SNN	CGS^{3}GCN
1	79.96	81.86	80.91	78.06	79.77	81.96	82.54	80.44	80.72	**82.83**
2	94.74	99.91	**100.00**	99.06	97.46	99.91	99.27	98.12	98.12	98.93
3	93.27	**100.00**	96.63	95.25	97.82	98.81	99.78	94.06	79.60	**100.00**
4	96.02	99.24	97.82	95.36	97.16	**99.81**	98.88	88.92	96.88	99.15
5	96.88	99.91	**100.00**	**100.00**	**100.00**	98.39	99.35	**100.00**	**100.00**	**100.00**
6	86.01	95.80	**100.00**	81.12	95.80	95.80	95.31	**100.00**	71.33	95.94
7	**95.24**	86.75	88.90	94.22	78.54	93.84	84.90	68.84	78.36	85.95
8	89.36	82.24	**95.82**	85.94	94.11	83.48	86.44	92.88	80.72	89.88
9	**90.56**	83.76	90.08	89.24	81.59	85.36	83.68	90.27	87.82	89.27
10	67.57	85.62	71.53	70.08	78.28	72.68	83.21	89.00	88.80	**94.94**
11	90.70	88.61	**99.81**	97.53	96.30	88.99	96.96	93.55	93.17	96.49
12	**96.45**	83.19	88.57	83.38	92.41	78.58	92.76	84.92	91.55	90.57
13	**91.58**	85.61	90.53	77.19	82.11	**91.58**	83.30	68.42	76.49	88.98
14	99.60	**100.00**	98.79	**100.00**	**100.00**	95.95	99.72	**100.00**	93.12	**100.00**
15	**100.00**	**100.00**	**100.00**	94.08	95.98	**100.00**	99.98	89.64	98.10	**100.00**
OA (%)	90.53	90.22	92.17	89.61	90.27	89.56	91.59	88.83	89.07	**93.47**
AA (%)	91.20	91.50	93.29	89.37	91.16	91.01	92.41	89.27	87.65	**94.20**
$\kappa \times 100$	89.72	89.38	91.50	88.72	89.44	88.67	90.87	87.87	88.14	**92.91**

TABLE X
CLASSIFICATION PERFORMANCE ON THE WHLK DATASET

Class NO.	MsET	MSST	DSFormer	MSFI-CNet	SFMamba	SSGRAM	MS2FN	AFSA	FRM-SNN	CGS^{3}GCN
1	94.77	96.51	**99.80**	96.82	99.49	96.37	97.80	98.02	95.56	99.74
2	89.28	85.15	**99.02**	86.62	**99.02**	98.12	94.47	96.92	95.64	98.02
3	94.21	91.38	**99.54**	96.71	97.84	94.10	98.23	99.33	91.70	97.65
4	87.01	81.93	87.55	**93.98**	91.89	84.95	91.57	92.92	86.45	92.14
5	82.89	84.18	93.87	61.66	97.94	**98.08**	90.18	84.38	83.22	94.86
6	95.82	92.24	99.30	99.01	99.24	98.93	98.79	96.67	95.69	**99.60**
7	96.81	99.97	99.65	97.93	99.83	**99.99**	99.96	94.74	99.88	99.95
8	76.03	74.96	83.65	77.30	87.17	90.99	89.98	35.11	**94.28**	92.75
9	90.63	79.70	93.02	86.90	84.66	84.17	90.85	**93.96**	84.89	91.81
OA (%)	91.89	90.95	95.04	94.42	96.36	93.75	95.92	92.73	93.56	**96.81**
AA (%)	89.72	87.33	95.04	88.55	95.23	93.97	94.65	88.00	91.92	**96.28**
$\kappa \times 100$	89.49	88.30	93.56	92.69	95.25	91.90	94.66	90.54	91.63	**95.83**

homogeneous regions due to the lack of explicit local spatial constraints. Frequency-domain methods (e.g., MSFI-CNet and SFMamba) effectively enhance spectral representations, yet still struggle in highly heterogeneous regions where long-range spatial–spectral dependencies are critical. Graph-based methods (e.g., SSGRAM and MS2FN) improve regional smoothness through neighborhood aggregation, but may over-smooth narrow boundaries and blend adjacent land-cover categories. Conventional SNN-based methods (e.g., AFSA and FRM-SNN) tend to generate patchy or incomplete structures because of the limited discriminative capability of sparse spike representations.

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TABLE XI
CLASSIFICATION PERFORMANCE ON THE MUUFL GULFPORT DATASET

Class NO.	MsET	MSST	DSFormer	MSFI-CNet	SFMamba	SSGRAM	MS2FN	AFSA	FRM-SNN	CGS^{3}GCN
1	91.04	90.06	90.74	91.07	90.29	90.33	91.62	87.72	87.99	**92.56**
2	86.84	91.45	86.24	83.91	83.22	87.29	91.65	79.00	82.88	**95.38**
3	85.56	86.02	83.78	83.15	82.59	84.24	86.92	77.29	79.47	**90.74**
4	96.06	95.62	97.67	95.17	94.64	95.21	97.04	90.97	91.84	**98.19**
5	91.16	92.27	85.42	89.76	90.59	92.33	89.74	86.02	86.20	**94.73**
6	99.62	98.42	**99.68**	99.37	99.30	99.43	99.11	98.73	98.61	98.61
7	94.08	93.24	94.34	94.62	94.43	93.95	**96.21**	92.13	91.61	95.58
8	94.08	94.78	95.06	95.36	**96.26**	92.62	95.93	89.92	91.65	96.18
9	89.64	84.32	80.08	84.01	87.81	91.26	90.25	83.38	78.41	**92.87**
10	93.94	99.39	90.91	92.42	93.33	96.97	98.79	92.73	92.12	**100.00**
11	99.16	**99.66**	99.16	99.16	98.82	99.16	**99.66**	99.16	98.99	99.33
OA (%)	90.69	90.73	89.51	89.99	89.80	90.22	91.68	86.00	86.82	**93.61**
AA (%)	92.83	93.20	91.19	91.64	91.94	92.98	94.27	88.82	89.07	**95.83**
$\kappa \times 100$	87.81	87.86	86.27	86.89	86.67	87.20	89.09	81.80	82.86	**91.60**

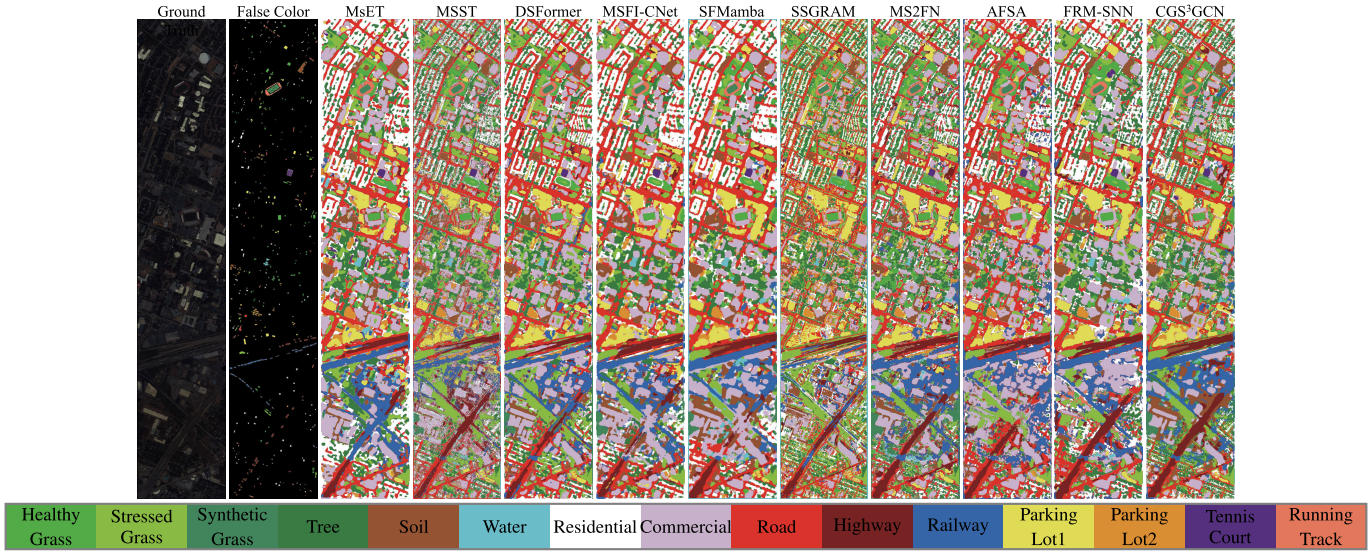


Fig. 11. Visual comparison of classification maps on the HU dataset.

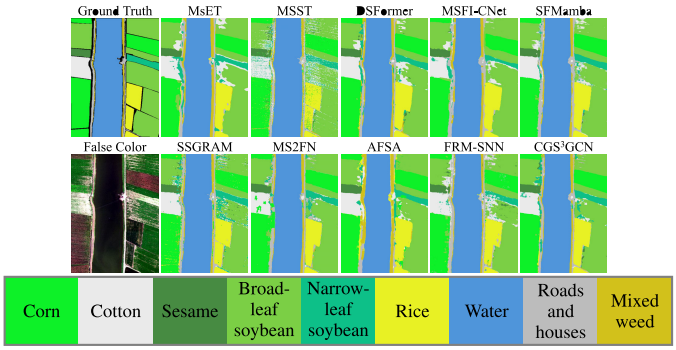


Fig. 12. Visual comparison of classification maps on the WHLK dataset.

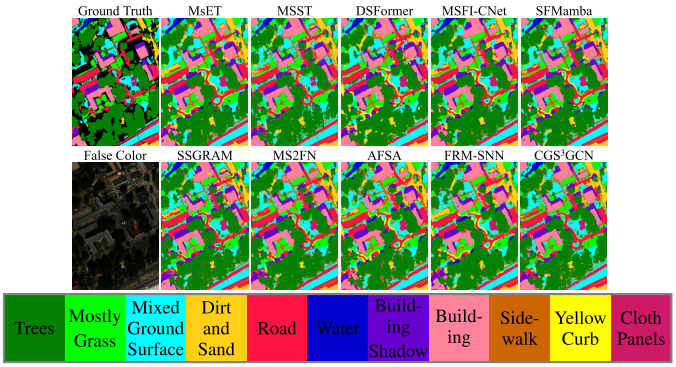


Fig. 13. Visual comparison of classification maps on the MUUFL Gulfport dataset.

Dataset-specific observations further reveal these limitations. On PU and WHLK, graph-based methods exhibit noticeable boundary over-smoothing around narrow structures such as roads and river edges, while Transformer-based methods produce scattered predictions in agricultural regions. On the more heterogeneous HU and MUUFL urban scenes, frequency-domain and SNN-based approaches show evident category mixing in residential and vegetation areas.

In contrast, CGS^{3}GCN generates more spatially coherent classification maps with clearer object boundaries and fewer misclassified regions. The proposed framework

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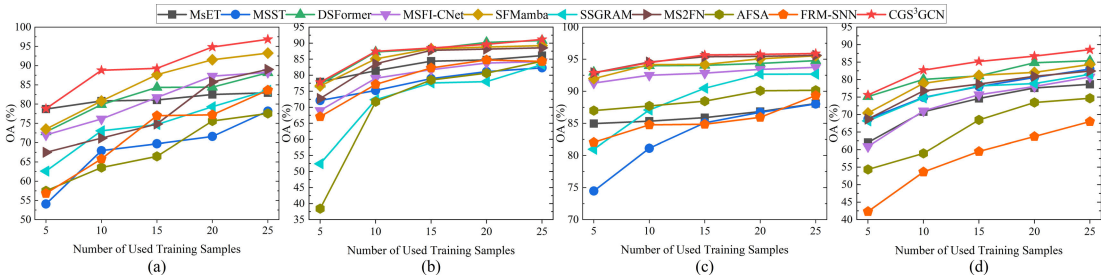


Fig. 14. OA (%) comparison under varying numbers of training samples on (a) PU, (b) HU, (c) WHLK, and (d) MUUFL datasets.

TABLE XII
COMPARISON OF COMPUTATIONAL COMPLEXITY AND INFERENCE EFFICIENCY

Models	Parameters (K)	Inference Memory (MB)	Inference Time (ms)
MsET	2292.21	732.28	43.48
MSST	422.58	**66.27**	17.50
DSFormer	676.52	237.42	14.16
MSFI-CNet	442.66	714.21	20.34
SFMamba	888.43	107.78	**6.17**
SSGRAM	2646.19	240.38	11.33
MS2FN	461.66	796.46	20.81
AFSA	2264.46	1110.61	12.52
FRM-SNN	1117.31	212.31	12.41
CGS^{3}GCN	**155.24**	156.59	10.49

effectively exploits complementary spatial, spectral, and temporal information, achieving a better balance between contextual aggregation and semantic preservation. Consequently, the predicted maps exhibit higher consistency with the underlying land-cover distributions and visual structures in the false-color images.

G. Computational Complexity Analysis

While the parameter count is solely determined by the network architecture, inference memory and runtime are additionally affected by the input scale. Therefore, to ensure a fair comparison, all methods are evaluated on the PU dataset using a standardized input consisting of a 15 $\times$ 15 spatial patch and 30 PCA-retained spectral bands. Inference memory and runtime are measured using a single forward pass with a batch size of 32.

As reported in Table XII, CGS^{3}GCN achieves the smallest parameter count (155.24k) among all compared methods. This compactness mainly stems from the proposed PCSEG, which performs spectral dependency modeling in a compact embedding space rather than directly operating on high-dimensional spectral features. In addition, STCA exploits lightweight temporal–channel recalibration based on spike statistics, avoiding the heavy computation associated with dense attention mechanisms.

Although the temporal dynamics of SNNs introduce additional memory consumption (156.59 MB) and runtime (10.49 ms), CGS^{3}GCN remains competitive in inference efficiency. Benefiting from the compact PCSEG, lightweight STCA, and shallow graph propagation enabled by CPG, the proposed framework achieves a favorable tradeoff between classification performance and computational cost. These results demonstrate that the performance gains of CGS^{3}GCN are achieved without relying on excessive model complexity.

H. Robustness to Limited Training Samples

To evaluate robustness under limited supervision, the number of training samples per class is varied from 5 to 25. The results are shown in Fig. 14.

As the number of training samples increases, all methods achieve improved performance. Nevertheless, CGS^{3}GCN consistently maintains competitive performance across different supervision levels, achieving the best results on PU and MUUFL and remaining comparable to the strongest competitors on HU and WHLK.

The narrowing advantage under very limited supervision can be attributed to two factors. First, with as few as five labeled samples per class, the fundamental information available for discrimination is scarce, which constrains all methods and reduces the absolute gap between them. Second, the proposed framework's prototype-conditioned spectral graph and CPG rely on learned features whose quality benefits from more training data, so their full potential is not realized under extreme scarcity. Nevertheless, CGS^{3}GCN remains competitive because the prototype statistics are computed directly from each input patch rather than learned from labels, and the sparse firing mechanism of SNNs provides inherent robustness that partially compensates for the reduced supervision.

These results demonstrate that the proposed framework can learn discriminative spatial–spectral representations with limited labeled samples and exhibits strong generalization capability under scarce-supervision scenarios.

I. Robustness to Noise

To evaluate robustness under noisy conditions, additive Gaussian noise and spectral perturbation with increasing disturbance levels are introduced into the test samples. The corresponding results are presented in Figs. 15 and 16, respectively.

As shown in Figs. 15 and 16, most competing methods exhibit noticeable performance degradation as the perturbation level increases. In contrast, CGS^{3}GCN consistently maintains competitive and stable performance across different datasets and noise levels, achieving the best or near-best results in most cases.

These results indicate that the proposed framework is robust to both additive and spectral perturbations, demonstrating the stability of the learned representations under noisy conditions.

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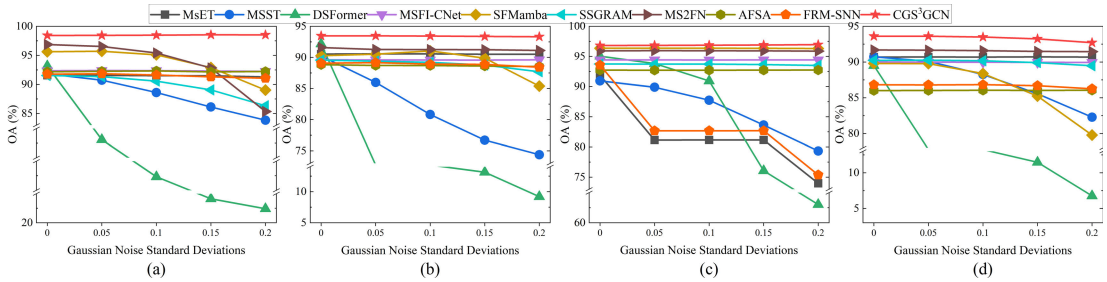


Fig. 15. OA (%) comparison under different Gaussian noise levels on (a) PU, (b) HU, (c) WHLK, and (d) MUUFL datasets.

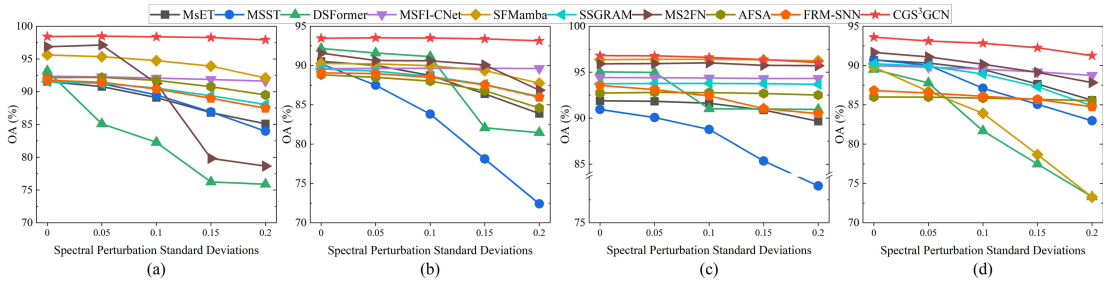


Fig. 16. OA (%) comparison under different spectral perturbation levels on (a) PU, (b) HU, (c) WHLK, and (d) MUUFL datasets.

V. Conclusion

In this article, we proposed a CGS^{3}GCN for HSIC. Specifically, a PCSEG was developed to construct compact and discriminative spectral node representations, an STCA module was introduced to exploit spike-based temporal–channel dependencies, and a CPG was designed to preserve target-oriented semantics during graph propagation. By integrating these components within a unified spatial–spectral spiking graph learning framework, CGS^{3}GCN achieves superior classification performance on four benchmark datasets and demonstrates strong robustness under limited training samples and noisy conditions.

Although promising results have been achieved, the current framework still has limitations that suggest promising future directions. First, the graph is built from Pearson-correlation affinities, which encode only linear pairwise relationships and are fixed independently of the classification objective; future work will explore task-driven adaptive graph evolution that learns richer, nonlinear graph structures jointly with the network. Second, the prototypes are limited to predefined center, mean, and max statistics, whose discriminative power is assumed rather than learned; future work will investigate learnable prototype generation optimized end-to-end for classification. Third, the static input is redundantly processed across T timesteps to induce the temporal dynamics that provide robustness, at a linear temporal cost; future work will pursue event-driven spiking graph architectures that preserve this robustness while reducing the redundancy for large-scale hyperspectral analysis.

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